

Program : Diploma in Robotic Process Automation	
Course Code : 5303B	Course Title: Non-Relational Database Management System.
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- Basics of database is studied already
- To develop basic concepts and appreciate the applications of database systems. Become familiar with the expanding role of database technology.
- To develop basic concepts and appreciate the applications of database systems. Understand the basic architecture and data models of a NoSQL database
- To understand Emerging features in the area of database security
- To demonstrate the basic concepts of transaction processing and concurrency control.

Course Prerequisites: Database Management Systems

Course Outcomes:

On completion of the course student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Will be able to identify different types of NoSQL Databases	12	Understanding
CO2	Demonstrate the features of indexing and hashing in database applications	17	Understand
CO3	To Demonstrate the basic concepts of transaction processing and concurrency control	15	Understand

CO4	Identify security and integrity threads in database	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3			2	3	3	2
CO2	3			2	3	3	3
CO3	3			2	3	3	2
CO4	3			3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Explain and compare different types of NoSQL Databases		
M1.01	NoSQL Data model: Aggregate Models- Document Data Model- Key	2	Understanding
M1.02	Key-Value Data Model Columnar Data Model	2	Understanding
M1.03	Graph Based Data Model	1	Understanding
M1.04	NoSQL system ways to handle big data problems	2	Understanding
M1.05	Moving Queries to data, not data to the query	1	Understanding

M1.06	Hash rings to distribute the data on clusters	1	Understanding
M1.07	Replication to scale reads	1	Understanding
M1.08	Database distributed queries to Data nodes.	2	Understanding
NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes.			
CO2	Demonstrate the features of indexing and hashing in database applications		
M2.01	Data Storage and Indexes- File organizations	2	Understanding.
M2.02	Primary indexing	3	Understanding
M2.03	Secondary index structures	2	Understanding
M2.04	Various index structures – hash-based	3	Applying
M2.05	Dynamic hashing techniques	3	Applying
M2.06	Multi-level indexes,	2	Applying
M2.07	B and B+ trees(insertion and deletion)	2	Applying
	Series Test – I	1	
Contents: Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion)			

CO3	To Demonstrate the basic concepts of transaction processing and concurrency control		
M3.01	Transaction concept, transaction state	2	Understanding
M3.02	Implementation of atomicity and durability	2	Understanding
M3.03	Concurrent executions	2	Applying
M3.04	Serializability	3	Applying
M3.05	Recoverability	2	Understanding
M3.06	Implementation of isolation	2	Understanding
M3.07	Testing for Serializability	2	Understanding
Contents: TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability.			
CO4	Identify security and integrity threads in database		
M4.01	Security and integrity threats, Defence mechanisms	2	Understanding
M4.02	Statistical database auditing & control	2	Understanding
M4.03	Security issue based on granting/revoking of privileges	2	Understanding
M4.04	Introduction to statistical database security	2	Understanding
M4.05	PL/SQL Security – Locks	2	Understanding
M4.06	Implicit locking, types and levels of locks, explicit locking	2	Understanding
M4.07	Oracles' named Exception Handlers	2	Understanding
	Series Test – II	1	

Contents:

Database security

Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers.

Text / Reference

T/R	Book Title/Author
R1	Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design and Application Programming, Pearson Education, 2013.
R2	Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015
R3	NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases and Big Data), Wiley, 2018
R4	Handbook of Database Security, Applications and Trends Gertz, Jajodia

Online Resources

Sl. No	Website Link
1	https://www.w3resource.com/redis/
2	https://www.w3schools.in/category/mongodb/
3	https://www.tutorialspoint.com/arangodb/index.html
4	https://www.tutorialspoint.com/cassandra/cassandra_introduction.html